

Leveraging Large Language Models to Locate Routers: Using LLMs for Parsing rDNS Hostnames

Danish Athar*
Purdue University
USA
danish8athar@gmail.com

Muhammad Abdullah*
Lahore University of
Management Sciences
Pakistan
27100242@lums.edu.pk

Kave Salamatian
Savoie-Mont Blanc
University
France
kave.salamatian@univ-
smb.fr

Frederick Douzet
University Paris 8
France
fdouzet@gmail.com

James Cowie
Independent Researcher
USA
jim.nh.us@gmail.com

Qasim Lone
RIPE NCC
Netherlands
qlone@ripe.net

Zafar Ayyub Qazi
Lahore University of
Management Sciences
Pakistan
zafar.qazi@lums.edu.pk

Zartash Uzmi
Lahore University of
Management Sciences
Pakistan
zartash@lums.edu.pk

Abstract

Extracting geographic hints from PTR hostnames is a common approach to geolocating Internet routers, but heterogeneous naming conventions across operators make this difficult. We present L⁴R, which treats PTR-hostname parsing as a semantic information extraction task. An LLM returns geographic hints, explanations, and granularity labels directly rather than first synthesizing extraction rules, a common approach in prior works. We evaluate this design with five complementary benchmarks which measure how many hints it extracts, how often they resolve to locations, and how often those locations agree with the references. On 500 Arelion PTR hostnames spanning 114 cities identified by operator mappings, the Gemini 3.7 Flash and Gemini 2.5 Flash configurations attain 99.8% and 100% end-to-end agreement. On Internet2, end-to-end agreement ranges from 93.5% to 97.8%. On a CAIDA RTT benchmark of 479 PTR hostnames from 263 domains, 51.1–66.8% of all targets are predicted within 40 km of their references. In a best-effort reconstruction of the Arelion portion of The Aleph’s Table 11, L⁴R with Gemini 3.7 produces 194 mapped hints and 138 distinct locations, compared with The Aleph’s published 58 mapped hints and 55 distinct locations. Direct semantic interpretation is therefore validated as a viable source of router-geolocation evidence.

1 Introduction

Geolocating Internet Protocol (IP) routers supports path analysis, failure planning [9], and detection of routes exposed to censorship [6]. Hostnames returned by reverse DNS pointer records (PTR hostnames) serve as useful evidence, for example the hostname `if-ae-49-2.tcore2.pvu-paris.as6453.net` points to Paris. Yet operators place geographic names, codes, and abbreviations in different positions, and no single convention covers them all [12].

PTR hostnames are operator-assigned and designed mainly for human interpretation. We therefore treat parsing as semantic information extraction rather than purely syntactic matching. Large language models can recognize geographic codes and use hostnames from the same domain as context. L⁴R turns these names into a geographic hint (geohint), explanation, and granularity label,

directly and without task-specific training or intermediate extraction rules.

We evaluate L⁴R using operator city mappings, locations inferred from round-trip-time (RTT) measurements, Internet exchange point (IXP) references, and a reconstruction of The Aleph’s Hoiho analysis. This is coupled with a qualitative analysis to demonstrate the mechanisms through which semantic interpretation finds cues that fixed patterns miss.

Our main contribution is a direct LLM approach to PTR-hostname interpretation and an analysis of where it succeeds and fails. We also assemble complementary benchmarks that distinguish hint extraction (yield), predicted location coverage, and agreement with reference locations. Since direct inference and compiled rules occupy complementary operating points but serve different needs, the results motivate a hybrid system that interprets uncovered names and caches or compiles validated mappings for reuse.

2 Background and Related Work

PTR hostnames have been used as a source of router-location evidence since operators often embed city names, airport codes, and facility identifiers into them. Their position, form, and meaning vary across and within an operator’s network, so prior systems combine hostname parsing with geographic dictionaries and measurements to interpret and validate them.

DRoP [4], Hoiho [8], and HLOC [10] explore different versions of this pipeline. Hoiho learns regular expressions from a collection of PTR hostnames in each domain, applies the rules to extract candidate hints, maps those hints through geographic dictionaries, and tests the locations using RTT measurements. HLOC matches hostname substrings against geographic databases, then uses active measurements to discard inconsistent candidates. Both depend on cues appearing in forms their rules or dictionaries express. Positional variation, combined tokens, non-standard abbreviations, and operator-specific codes can therefore reduce coverage before validation even begins. Section 4.6 illustrates these cases.

Thiagarajan et al. introduced The Aleph in a preliminary poster [11] and subsequent paper [12]. Its three-stage, per-operator pipeline groups PTR hostnames by structural pattern, generates an extraction regex for each class, and uses an LLM to map extracted hints

*Danish Athar and Muhammad Abdullah contributed equally to this work.

to locations. Once generated, these artifacts can be applied deterministically and amortized across large hostname collections. However, extraction remains bounded by the generated rules because of which adding a network or adapting to changed conventions requires rerunning its training pipeline.

L⁴R instead interprets batches of hostnames from the same domain using a fixed prompt, without per-domain training, retraining, or rule synthesis. The Aleph’s results already show that LLMs can map custom hints to locations [12]. We examine whether direct interpretation can also replace regex generation and extraction.

3 Methodology

3.1 Direct PTR-Hostname Interpretation

We group PTR hostnames by naming domain so the LLM can use the broader naming pattern used by an operator as context. The prompts define city, region, country, airport, United Nations location codes (UN/LOCODEs), and Common Language Location Identifier (CLLI) codes. They also allow unfamiliar, operator specific codes to be inferred from within-domain context. For each hostname, the model returns a geohint containing a geographic name or code, an explanation of its interpretation, and a granularity label such as city, region, or country. The explanations help us inspect the predictions.

All runs use the prompts presented in Appendix C. Each request contains only the naming domain and PTR hostnames. The model receives no IP addresses, autonomous system (AS) numbers, router identifiers, reference locations, and has no access to external tools. We use Gemini 3.7 Flash as the primary configuration, Gemini 2.5 Flash to test an earlier model, and Gemma 4 26B A4B as an open-weight baseline. Inputs are ordered deterministically between runs. Provider defaults are retained, including Gemini’s dynamic thinking for the Flash models. The code and data used for the pipeline and benchmarks are available in Appendix B.

3.2 Batching

A batch refers to the set of PTR hostnames grouped under a domain, sent per LLM call. The size of a batch is a hyperparameter in L⁴R varying which trades wider within-domain context for shorter outputs. The Flash configurations use batches of up to 500 PTR hostnames. Gemma due to its shorter context window is limited to 50-names per batch. It was also noticed for larger batches that responses start to omit domains, so batch size must be tuned on a per model basis.

3.3 Benchmarks and Reference Evidence

Table 1 summarizes the five benchmarks. In each benchmark, L⁴R first produces a geographic hint. A benchmark-specific resolver then converts that hint into a city or coordinates for comparison with the withheld reference. The reference evidence differs between benchmarks. The operator benchmarks test agreement with operator-published city mappings. The CAIDA RTT tests proximity to locations inferred from latency measurements. The IXP benchmark derives an indirect city reference from an IXP-associated interface on the same router. The Aleph extension counts extracted hints and mapped locations. These benchmarks thus measure different aspects of the approach.

Arelion City Mapping. This benchmark contains 500 unique PTR hostnames across 114 cities and 35 countries. We join site codes to Arelion’s Origin+ city prefixes [1], deduplicate devices, and cap sampling at 15 hostnames per city. We also average results with equal weight per city and evaluate one target per city to check whether repeated cities dominate the results.

Internet2 PoP Location. The 46 PTR hostnames represent 46 point-of-presence (PoP) codes and 41 US cities. References come from directly connected measurement endpoints and operator-published mappings exposed through Internet2’s Looking Glass [5]. They are assigned independently of the hostname being evaluated.

CAIDA RTT Proximity. CAIDA reports 4,838 interfaces located through RTT proximity to RIPE Atlas probes [2]. We retain 479 historical PTR hostnames that could be recovered, covering 263 domains and 42 countries. These correspond to 496 interface records and 185 reference coordinates. A prediction agrees with the reference if it lies within 40 km of the nearest reference point associated with that hostname.

IXP-Anchored Router Recovery. This benchmark consists of routers from the February 2024 ITDK assigned a location through IXP evidence after Hoiho abstains [3]. For each router, an IXP-associated interface provides the city reference, while L⁴R sees only PTR hostnames from different, non-IXP interfaces. The benchmark contains 397 routers, 765 distinct PTR hostnames, 278 naming domains, and 51 cities in 36 countries. PTR predictions are aggregated at the router level. This deliberately difficult set tests whether L⁴R can recover location evidence where Hoiho fails to recover a location. It does not estimate performance on routers in general.

Aleph Table 11: Arelion Extension. The Aleph reports extraction and mapping counts (yield) for seven ITDK networks [12]. We reconstruct the Arelion (AS1299) portion as closely as public data allow, using 6,385 PTR hostnames from 59 domains in the February 2024 ITDK. A second LLM stage maps extracted hints to localities. We count distinct hints, mapped hints, and distinct locations.

3.4 Scoring

Let N be the number of targets, P the number with a resolved location prediction, and C the number whose prediction agrees with the reference. Prediction coverage (Cov.) is P/N : how often the approach produces a usable location. Conditional agreement (Cond.) is C/P : how often a resolved prediction agrees with the reference. End-to-end agreement (E2E) is C/N : the fraction of all targets with an agreeing prediction, counting missing and unresolved outputs as failures. The target is a PTR hostname, except in IXP recovery, where it is a router.

Arelion and Internet2 score only the raw geohint against code-to-city mappings. Their strict operator dictionaries permit this direct comparison. CAIDA RTT, by contrast, requires conversion of a free-form location description into coordinates, so its resolver also sees the explanation and granularity, but neither the PTR hostname nor the reference.

For IXP recovery, we retain a fixed resolver that does not see the reference answers. It leaves many free-form aliases and facility names unresolved. We therefore also report a semantic review that interprets the model’s intended location with the reference answers visible. We treat this review as exploratory, not as blinded accuracy.

Table 1: Evaluation benchmarks and roles. References are withheld from extraction.

Dataset	Unit	Reference	Role
Arelion City Mapping	PTR hostname	Operator Origin+ city codes	City-code agreement
Internet2 PoP Location	PTR hostname	Operator PoP mappings	PoP-city agreement
CAIDA RTT Proximity	PTR hostname	RTT proximity to probes	Distance agreement
IXP-Anchored Router Recovery	Router	IXP anchor on separate interface	Hoiho-abstention stress test
Aleph Table 11: Arelion Extension	Hint / locality	Published Table 11 counts	Historical yield

4 Analysis

4.1 Arelion City Mapping

Table 2: Arelion operator-city agreement (500 PTR hostnames; 114 cities).

System	Cov.	Cond.	E2E	1/city
Flash 3.7	499/500	100%	99.8%	114/114
Flash 2.5	500/500	100%	100%	114/114
Gemma 4	417/500	100%	83.4%	95/114

Every resolved L^4R prediction agrees with the operator city (Table 2), so coverage and end-to-end agreement are identical. Flash 3.7, Flash 2.5, and Gemma resolve 499/500, 500/500, and 417/500 hostnames, respectively. When evaluated on one deterministically selected hostname per city, the Flash configurations cover all 114 cities and Gemma covers 95. The results remain similar when cities with more hostnames receive no extra weight.

We also queried The Aleph’s public API on these 500 PTR hostnames. Each query contained only the PTR hostname and ASN 1299. The API used the Aleph’s historically generated deployed rules and was not retrained on this benchmark. All requests completed, but none returned a regex, geohint, or city, yielding prediction coverage of 0/500. AS1299 has two deployed regexes for `-bN-link` and `-ic-` forms. They do not match benchmark forms such as `adm-b14`. This shows how deployed rule coverage can fall as operator forms change. Direct interpretation requires no rule refresh. The Aleph’s published validation [12] found 38 mapped hints among 95 Arelion Looking Glass mappings, where all 38 were correct, showing useful coverage for the naming forms evaluated then. Regenerating the rules on newer PTR hostnames could restore coverage for the new forms while L^4R covers the observed forms directly.

4.2 Internet2 PoP Location

Table 3: Internet2 PoP-city agreement (46 PTR hostnames).

Model	Cov.	Cond.	E2E
Flash 3.7	93.5%	100.0%	43/46 (93.5%)
Flash 2.5	97.8%	100.0%	45/46 (97.8%)
Gemma 4	93.5%	100.0%	43/46 (93.5%)

Every prediction resolved by the operator code dictionary agrees with the reference city (Table 3), yielding perfect conditional agreement. The remaining hints are not accepted by the conservative, fixed resolver. For example, it does not accept `newy` for the reference code `newy32aoa`, even when the explanation names New York. The two-hostname difference between the Flash configurations is not statistically distinguishable (two-sided exact McNemar

test, $p = 0.5$). The main observation across this and the Arelion City Mapping benchmark is the robustness seen across models on structured operator nomenclature.

4.3 CAIDA RTT Proximity

Table 4: CAIDA RTT agreement within 40 km (479 PTR hostnames).

Model	Cov.	Cond.	E2E
Flash 3.7	72.9%	91.7%	320/479 (66.8%)
Flash 2.5	59.9%	90.9%	261/479 (54.5%)
Gemma 4	60.8%	84.2%	245/479 (51.1%)

Results vary more across models on CAIDA’s heterogeneous naming domains (Table 4). The counts for predicted locations within the 40 km are 320, 261, and 245, for Flash 3.7, Flash 2.5 and Gemma respectively. Median errors are similar at 3.77 km for both Flash configurations and 4.16 km for Gemma. Gemma returns 424 nonempty hints, compared with 391 for Flash 3.7 and 362 for Flash 2.5, but produces the fewest locations within 40 km of the reference (245). Extracting more hints therefore does not necessarily correlate with better geolocation accuracy.

4.4 IXP-Anchored Router Recovery

Table 5: IXP recovery with intent aware resolver (%).

Model	Cov.	Cond.	E2E
Flash 3.7	46.9	81.2	38.0
Flash 2.5	35.5	75.9	27.0
Gemma 4	30.0	59.7	17.9

Flash 3.7 returns nonempty hints for 346/765 PTR hostnames, but the frozen resolver converts them into city predictions for only 19/397 routers. Twelve predictions agree with the IXP-derived references, giving 4.8% coverage, 63.2% conditional agreement, and 3.0% E2E. The large gap between extracted hints and resolved cities indicates that normalization is the main bottleneck in this cohort, rather than the model itself.

To explore this we conduct intent-aware resolution which recovers substantially more usable predictions (Table 5). Across models, coverage ranges from 30.0% to 46.9%, and 59.7–81.2% of resolved predictions agree with the IXP-derived city. Most routers remain unresolved, but the review identifies plausible locations where Hoiho provides none. Because the resolver sees the reference answers, these results are exploratory, but they show that direct interpretation can extend coverage into naming domains where rule-based extraction fails.

4.5 Aleph Table 11: Arelion Extension

Table 6: Aleph Table 11 extension: hint yield.

Method	Distinct Hints	Mapped Hints	Locations
Hoiho (published)	9	9	9
Aleph (published)	58	58	55
Flash 3.7	274	194	138
Flash 2.5	276	167	123
Gemma 4	305	174	133

Table 6 compares extraction and mapping counts with The Aleph’s published results. Flash 3.7 produces 274 distinct hints, of which 194 map to 138 locations. The published Aleph counts are 58 mapped hints and 55 locations. Gemma produces more distinct hints (305), but fewer mapped hints and locations than Flash 3.7. All three L⁴R configurations exceed the published counts.

Flash 3.7, Flash 2.5, and Gemma return nonempty hints for 12.51%, 13.47%, and 15.11% of PTR hostnames, respectively. Hints mapped to locations account for 8.47%, 7.77%, and 8.39% of inputs. Unlike the curated operator benchmark, this AS-level corpus includes customer and other naming domains. Though it should be noted that these yield counts measure extraction and mapping output, not accuracy.

4.6 Qualitative Analysis

To understand how direct interpretation differs from rule-based extraction, we conducted an experiment using Gemini 2.5 Flash on approximately 35,000 PTR hostnames from 302 domains in Hoiho’s data supplement [7]. Flash 2.5 produces 5,462 hints, compared with 3,179 in Hoiho’s accepted hints list. Manual review of the hostname examples and model explanations revealed several recurring differences in how geographic cues are recovered.

Positional variation. Geographic cues need not occupy consistent positions, even within one naming domain. Hoiho’s `acorus.net` pattern extracts `nyc`, `par`, and `lis`, but misses `seattle` at the beginning of `seattle-ix.wes.acorus.net`. The city name is explicit, but falls outside the positions covered by the learned pattern. Direct interpretation recognizes it without requiring a separate rule for that hostname structure.

Distributed and within-domain evidence. Geographic information may be distributed across multiple hostname segments. In `altiusbb.net`, Hoiho’s regex extracts locality names such as `anderson`, `cherrygrove`, and `yoder`, but misses `liberty` and `el-jobean`. In contrast, L⁴R interprets `liberty-core-02.in.altiusbb.net` as `Liberty, Indiana`, combining the locality with the state abbreviation `in`. It similarly identifies `El Jobean, Florida`, from `el-jobean-bhrtr.fl.altiusbb.net`. Other hostnames in the domain can also provide context. Both methods recover `tmk` in `zsttk.ru`. L⁴R’s explanation links it to `Tomsk` by referring to names containing `tomsksoft` in the same domain. Here, the difference is in explaining the extracted code, rather than finding an additional hint.

Operator-specific identifiers. Some operators use facility names and administrative identifiers alongside standard geographic codes. In `cybermesa.com`, Hoiho recognizes codes such as `snfenm`, but misses `simms` in `ms1.simms.cybermesa.com`. The model interprets `simms` as a reference to the `Simms Building` in `Albuquerque`. Likewise, it recognizes `cesa06` and `cesa09` in `wiscnet.net` as `Wisconsin`

Cooperative Educational Service Agency regions. Interpreting these identifiers requires knowledge of local buildings or organizations.

Language and alternate forms. In `cyberlink.ch`, Hoiho recovers the standard airport code `zrh`, but its regex misses `geneve` and `zuerich`. L⁴R identifies these in `gw-gutz-geneve2-fb.cyberlink.ch` and `gw-rheiair-zuerich.cyberlink.ch`, interpreting `geneve` as `Geneva` and `zuerich` as `Zurich`. In `space.net`, Hoiho’s regex extracts `muc` and `muenchen`, but neither appears in its accepted hint list. L⁴R interprets them as `Munich Airport` and `Munich`, respectively. These examples show how the model handles both alternate spellings and abbreviated names.

Interpreting unaccepted regex matches. Some geographic strings are captured by Hoiho’s regex but excluded from its accepted hint list. For `celeritybroadband.com`, these include `mtmrrs`, `erwin`, and `ftrp`. L⁴R also recovers these strings and supplies explanations for their geographic meaning. For example, it returns `mtmrrs` from `mtmrrs-rb1100-e3.celeritybroadband.com` and interprets the abbreviation as `Mount Morris, Illinois`. Here, the regex already finds the relevant token. The added value lies in interpreting it rather than extending pattern coverage. This suggests that semantic interpretation can also complement existing extraction rules by proposing mappings for otherwise unrecognized hints.

Together, these examples show how direct interpretation broadens hint recovery through flexible token placement, combined evidence, and geographic knowledge.

5 Discussion and Conclusion

Across the five benchmarks, direct interpretation recovers geographic information without first generating extraction rules. The harder task is resolving and validating that information. The strong Flash models recover nearly all locations in the two structured operator corpora, whereas coverage varies substantially across the heterogeneous RTT and IXP benchmarks. More extracted hints also do not imply more correct locations: Gemma emits the most nonempty RTT hints (424) but produces the fewest correct locations (245) on the RTT benchmark. The IXP benchmark shows the same bottleneck from another direction, as many plausible hints cannot be normalized by a fixed resolver.

Direct inference and compiled rules therefore occupy complementary operating points. Rules amortize well over repeated, billion-scale scans of stable naming conventions. Direct interpretation is better suited to router-scale collections, unseen domains, and the long tail of PTR hostnames that existing rules do not cover. It can also accommodate changing operator nomenclature without first regenerating extraction rules.

Thus a hybrid system that invokes direct interpretation for unknown or low-confidence PTR hostnames, validates the resulting locations using operator data, topology, or active measurements, and caches or compiles stable mappings for deterministic reuse may offer a better balance between accuracy and scalability. Future work should determine when such mappings are reliable enough to reuse, how validation should be triggered, and whether this design improves coverage without sacrificing accuracy or operational efficiency. Overall, our results show that direct semantic extraction is a viable source of router-location evidence, while reliable validation and reuse remain the central challenges.

References

- [1] Arelion. 2026. Twelve99/AS1299 BGP communities. <https://www.arelion.com/our-network/bgp-routing/bgp-communities> Origin + communities section; last updated May 26, 2026.
- [2] CAIDA. 2016. CAIDA UCSD Geolocated Router Dataset. doi:10.23721/1393258
- [3] CAIDA. 2024. The CAIDA Macroscopic Internet Topology Data Kit – 2024-02. <https://www.caida.org/catalog/datasets/internet-topology-data-kit>
- [4] Bradley Huffaker, Marina Fomenkov, and kc claffy. 2014. DRoP: DNS-based router positioning. *SIGCOMM Comput. Commun. Rev.* 44, 3 (July 2014), 5–13. doi:10.1145/2656877.2656879
- [5] Internet2. n.d. Looking Glass. <https://docs.periscope.ns.internet2.edu/looking-glass/> Periscope documentation; accessed September 6, 2026.
- [6] Josh Karlin, Stephanie Forrest, and Jennifer Rexford. 2009. Nation-State Routing: Censorship, Wiretapping, and BGP. arXiv:0903.3218 [cs.NI] <https://arxiv.org/abs/0903.3218>
- [7] Matthew Luckie, Bradley Huffaker, Alexander Marder, Zachary Bischof, Marianne Fletcher, and k claffy. 2021. Data supplement for “Learning to Extract Geographic Information from Internet Router Hostnames”. <https://publicdata.caida.org/datasets/supplement/2021-conext-hoiho/>
- [8] Matthew Luckie, Bradley Huffaker, Alexander Marder, Zachary Bischof, Marianne Fletcher, and k claffy. 2021. Learning to Extract Geographic Information from Internet Router Hostnames. In *Proceedings of the 17th International Conference on Emerging Networking EXperiments and Technologies* (Virtual Event, Germany) (CoNEXT '21). Association for Computing Machinery, New York, NY, USA, 440–453. doi:10.1145/3485983.3494869
- [9] Ramakrishna Padmanabhan, Aaron Schulman, Dave Levin, and Neil Spring. 2019. Residential Links Under the Weather. In *Proceedings of the ACM Special Interest Group on Data Communication* (Beijing, China) (SIGCOMM '19). Association for Computing Machinery, New York, NY, USA, 145–158. doi:10.1145/3341302.3342084
- [10] Quirin Scheitle, Oliver Gasser, Patrick Sattler, and Georg Carle. 2017. HLOC: Hints-Based Geolocation Leveraging Multiple Measurement Frameworks. In *2017 Network Traffic Measurement and Analysis Conference (TMA)* (Dublin, Ireland). IEEE, 1–9. doi:10.23919/TMA.2017.8002903
- [11] Kedar Thiagarajan, Esteban Carisimo, and Fabián E. Bustamante. 2024. Poster: Revealing Hidden Secrets: Decoding DNS PTR records with Large Language Models. In *Proceedings of the ACM SIGCOMM 2024 Conference: Posters and Demos* (Sydney, NSW, Australia). Association for Computing Machinery, New York, NY, USA, 10–12. doi:10.1145/3672202.3673717
- [12] Kedar Thiagarajan, Esteban Carisimo, and Fabián E. Bustamante. 2025. The Aleph (N): Decoding Geographic Information from DNS PTR Records Using Large Language Models. *Proc. ACM Netw.* 3, CoNEXT1, Article 7 (March 2025), 20 pages. doi:10.1145/3709374

A Acknowledgments

We thank Antonin Rosa-Martin for his valuable contributions to our discussions and for sharing data that supported this work. We also thank Isabelle Saint-Mézard for her thoughtful contributions to our discussions and Nowmay Opalinski for helping establish the research collaboration that made this work possible.

The project was Co-funded by the European Union (ERC DATAROUTES #101098032). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.

B Artifact Availability

The artifact is available at [this repository](#). It contains the L⁴R prompts and model configurations, model-visible benchmark inputs, construction and scoring code, accepted model responses and consolidated predictions.

C L⁴R Prompts

Following are the system and task prompts used for all benchmark runs. Each request substitutes the domain and hostname list into the placeholders. JSON braces are reproduced exactly as sent. The artifact includes the exact prompt strings and Python template.

System prompt

You are an expert network analyst specializing in reverse DNS hostname interpretation and geolocation analysis.

Your expertise includes:

- Understanding network infrastructure naming conventions
- Recognizing geographic codes in various formats (IATA, ISO, CLLI, UN/LOCODE) and unique naming conventions
- Distinguishing between genuine geographic hints and technical/service identifiers
- Analyzing patterns across multiple hostnames within a domain

Always return valid, structured JSON output that can be directly parsed for further analysis.

Task prompt

For the given domain, analyze each hostname and extract geolocation hints (geohints) embedded within it.

```
<Geohint Definition>
Valid geohints include:
- City/region codes (2-5 letters, e.g., "lax",
"fra")
- Country codes (ISO 2-letter, e.g., "us",
"de")
```

- Airport codes (3-letter, e.g., "JFK", "CDG")
- UN/LOCODEs (5 characters, e.g., "USNYC")
- CLLI codes (several characters)
- Recognized city/region names (e.g., "tokyo", "sofia")
- Possible shortened/coded names for physical locations
 - For unrecognized codes or words, you may also try to infer the possible location it represents by looking at the other hostnames for the same domain, but only accept the geohint if it makes sense

```
</Geohint Definition>
```

```
<Domain Name>
```

```
{domain_name}
```

```
</Domain Name>
```

```
<Hostname data>
```

```
{hostname_data}
```

```
</Hostname data>
```

Return structured JSON in the following format:

```
<Output Structure>
```

```
{
  "domain": "example.com",
  "hostnames": {
    "nyc1-edge.example.com": {
      "geohint": "nyc",
      "explanation": "IATA code for New York City",
      "granularity": "city"
    },
    "lon-gw01.example.com": {
      "geohint": "gw",
      "explanation": "GW likely refers to Gatwick
Airport in London since Lon is present in the
hostname which could mean London",
      "granularity": "airport"
    },
    "router-edge-klmzmi01-usa.example.com": {
      "geohint": "klmzmi",
      "explanation": "CLLI code for Kalamazoo,
Michigan. Makes sense given the USA mention as
well.",
      "granularity": "area/region"
    }
  }
}
```

```
</Output Structure>
```

Use the JSON format from the example. Do NOT add markdown formatting or extra characters.

D Ethics and AI Use

The study analyzes infrastructure hostnames rather than user communications. Hosted inference sends these hostnames to model providers, so deployments should follow confidentiality requirements and the source datasets' terms of use. LLMs perform extraction, mapping and the semantic review described in the paper. LLMs assisted with coding and manuscript preparation.